A Knowledge-Guided Bi-Modal Network for the Classification of Anterior Chamber Angle Images-supplementary

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1. Summary of Notations

Table S1 summarizes the main notations and symbols used throughout this paper for clarity and reference.

Table S1 Notation summary.

Symbol	Description
X	A training samples
x_I	Input ACA image of sample I, $x_I \in \mathbb{R}^{3\times H\times W}$
y_I^c	Image-level label (category-level label) of sample I
y_I^s	Pixel-level label of sample I (available for partially annotated data)
t^c	Domain knowledge (text description) for category c
v^{N2}	Sequential encoding vector for ACA level N2
$\mathcal{T}$	Domain knowledge for all ACA levels (textual descriptions)
$\mathcal{V}$	Set of encoded domain knowledge vectors, $\mathcal{V} \in \mathbb{R}^{N_c\times A}$
N_c	Number of ACA categories (e.g., N1-N4, W)
A	Number of attributes per ACA level (e.g., 5)
$\mathcal{C}$	Set of five ACA image categories: N1,N2,N3,N4,W
$P_{subscript}$	The hidden features in different layers of the model

2. Experimental setup

we use five different metrics to measure classification results from the model: Accuracy (ACC), Sensitivity (SEN), Precision (PR), the area under the receiver operating characteristic curve (AUC), and F1-score (F1), where ACC is the primary metric for assessing the prediction accuracy of the models.

These metrics are defined as follows:

[1] These authors contributed equally to this work.

$$\mathrm{Acc} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}$$

$$\mathrm{SEN} = \frac{TP}{TP + FN}$$

$$\mathrm{PR} = \frac{TP}{TP + FP}$$

$$\mathrm{F1} = \frac{2 \times PR \times SEN}{PR + SEN}$$

where TP, FP, TN and FN denote the numbers of true positives, false positives, true negatives, and false negatives, respectively.

3. Independent Discriminative Capacity of the Knowledge Module

To quantify the independent discriminative capacity of the knowledge embedding, we implemented a retrieval-style baseline in which each test sample was classified by computing the cosine similarity between its predicted knowledge vector and all predefined class knowledge prototypes, with the class of highest similarity assigned as the prediction. As shown in Fig S1, at full coverage (100% of test samples), the knowledge-only retrieval achieved an accuracy (ACC) of 0.7208 and an area under the ROC curve (AUC) of 0.7704, indicating that the knowledge embeddings alone encode substantial class-discriminative information. We further evaluated performance under varying confidence thresholds, finding that as coverage decreases, both ACC and AUC generally improve. Specifically, at 68.5% coverage, ACC reaches 0.7852 and AUC 0.7855, and at 57.4% coverage, ACC further increases to 0.7876. This monotonic improvement demonstrates that high-confidence samples are well separated in the knowledge embedding space, whereas lower-confidence samples mainly contribute uncertainty rather than arbitrary misclassification. These results directly support the notion that the knowledge module independently captures meaningful semantic alignment and provides non-trivial discriminative information. For reproducibility, the evaluation was performed without training or fine-tuning on the test set, using consistent random seed 42, with performance statistics computed across both the full test set and confidence-thresholded subsets.

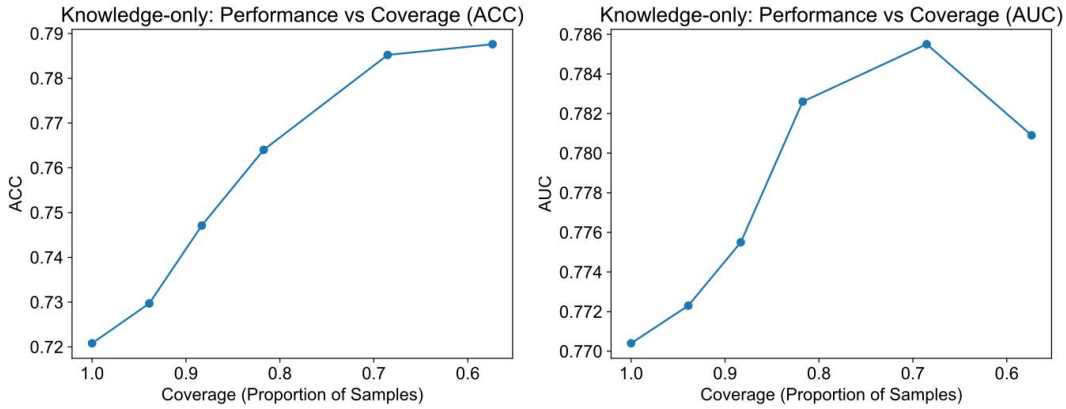

Fig.S1 Performance of the knowledge-only retrieval baseline under varying coverage levels. Left: Accuracy (ACC) versus coverage, where coverage denotes the proportion of test samples retained by descending maximum prediction confidence. ACC increases as coverage decreases, indicating that high-confidence samples exhibit stronger class separability. Right: Area under the ROC curve (AUC) versus coverage, reflecting overall ranking ability. AUC initially increases with decreasing coverage, peaking around 68% coverage, and slightly decreases at very low coverage due to reduced sample statistics. These trends demonstrate that the knowledge embeddings independently capture meaningful semantic information: high-confidence samples are well separated, while lower-confidence samples mainly contribute uncertainty rather than arbitrary errors, supporting the discriminative value of the knowledge module.

4. Adaptation and Implementation Details of MedCLIP for ACA Classification

To ensure a fair comparison and full reproducibility, we provide detailed information on how MedCLIP was adapted to our ACA classification task.

First, regarding pretrained weights, the text encoder was initialized with Bio_ClinicalBERT weights, following standard practice for medical-language modeling. The visual encoder used the default MedCLIP ResNet-50 pretrained weights without any architectural modifications.

Second, the image-text pairing strategy was adjusted to accommodate our task. Unlike MedCLIP's original design, which relies on large-scale image-level captions, our task uses class-level medical knowledge. Each image is paired with a single textual description corresponding to its class, and this description is shared among all images of the same ACA grade. In other words, all images

belonging to a given class use identical class-level textual descriptions that encode domain knowledge associated with that grade. This design is consistent with the problem setting and is explicitly stated in the manuscript: "Specifically, while MedCLIP is designed for image-text contrastive learning using large-scale image-level captions, our approach relies on class-level textual descriptions, i.e., class-text pairs, which reflect domain knowledge associated with each ACA grade."

Third, the training configuration was as follows: batch size 16, 10 epochs, optimizer AdamW, learning rate 2×10^{-5}, weight decay 1×10^{-4}, and a warm-up ratio of 0.1. All other hyperparameters followed the original MedCLIP implementation to avoid introducing confounding factors.

Finally, to ensure a fair evaluation, MedCLIP was trained and tested using the same data split as our proposed method, with a random 80% / 20% division for training and testing, respectively. To further improve reproducibility, replicable training scripts and configuration files have been uploaded to a public GitHub repository (https://github.com/Stardus-t/MedClip_.git), with the corresponding commit hash b30dd309f94c8475b4c7a76f1c186579adcb8742.

5. Cross-Task Generalization on Diabetic Retinopathy Classification

We further validate the generalizability of KGNet on diabetic retinopathy classification using structured clinical knowledge. The results demonstrate consistent performance gains over diverse baselines, highlighting the effectiveness of knowledge-guided bimodal learning in complex retinal image analysis tasks.

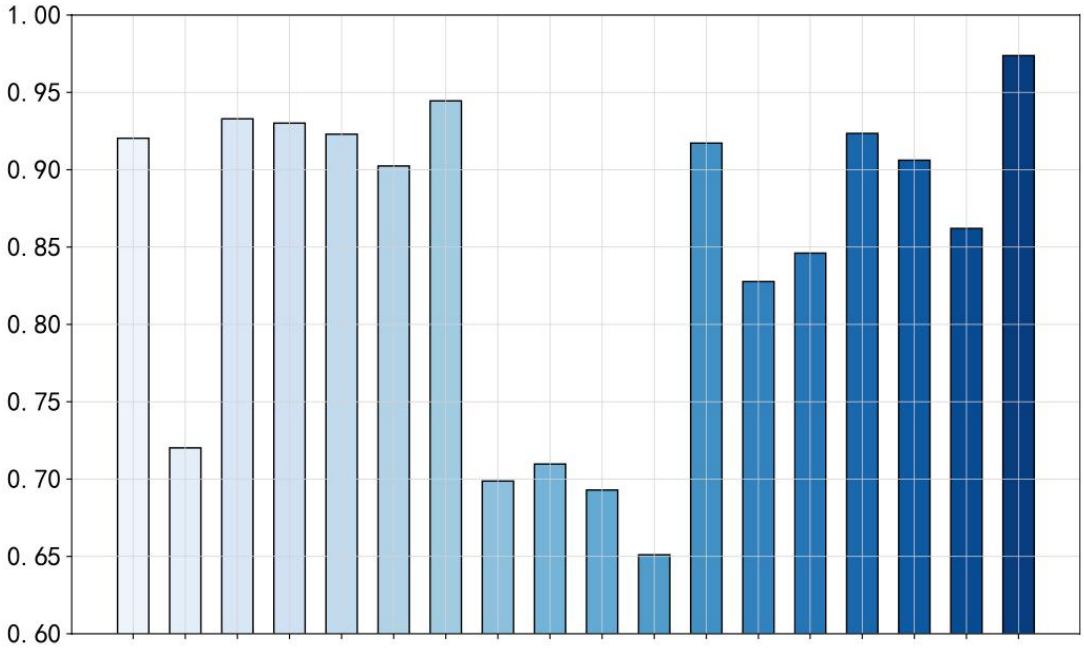

Fig S2. Comparison of AUC scores across different models on the DR classification task. KGNet achieves the highest AUC, demonstrating its superior discriminative ability through domain knowledge integration. The abscissas are Vgg, GoogLeNet, EfficientNet, DenseNet, ResNet, SEResNet, ECAResNet, TinyViT, FastViT, ConViT, MLP-Mixer, FixMatch, CCT, UPS, RIRResNet, RCRResNet, FITNet and KGNet respectively.

6. Computational Cost Evaluation Protocol

To ensure reproducibility and enable fair comparison of computational costs, we provide a detailed description of the measurement protocol used for reporting FLOPs, parameters, and inference time. All experiments were conducted on a single NVIDIA GPU (model specified in the manuscript). FLOPs and inference time were measured using an input tensor of size (1, 3, 128, 256) with batch size 1, reflecting the standard evaluation setting used throughout our experiments. The model's MACs (Multiply-Accumulate operations) and parameter counts were computed using the THOP profiling tool while the model was in evaluation mode (i.e., model.eval()). For inference time measurement, the model was first switched to evaluation mode, followed by 10 warm-up iterations to stabilize GPU execution. Inference time was then averaged over 300 forward passes, with the reported latency corresponding to the mean per-image inference time.

Table S2.Computational efficiency comparison of different models in terms of FLOPs (G), number of parameters (M), and inference time (ms).

Model	**Flops/G**	**Params/M**	**Times/ms**
VGG	5.02	18.93	1.18
GoogLeNet	0.84	22.33	8.96
EfficientNet	0.12	3.97	7.36
Densenet	0.93	6.88	12.92
ResNet	1.35	24.05	5.10
SEResNet	1.35	26.05	7.40
ECAResNet	1.43	23.54	7.37
TinyViT	2.04	20.61	7.67
FastViT	3.00	42.78	59.34
ConViT	0.54	5.47	7.63
MLP-Mixer	6.31	59.12	3.89
FixMatch	192.98	87.76	8.65
CCT	23.14	46.85	4.94
UPS	192.98	87.76	8.66
RIRResNet	1.36	29.13	10.86
RCRResNet	1.35	24.08	17.78
FIT-Net	2.20	94.97	26.7222
KGNet	293.21	125.10	13.30